



Integration · Worksheet 1

Chapter 8: Integration. Answer all questions. Show your working clearly. Total: 220 marks. Time: 4 hours 35 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are ten sections, one for each topic in Chapter 8. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Remember $+C$ on every indefinite integral. All angles are in radians.
- When a question asks for an exact answer, solutions relying entirely on calculator technology are not acceptable.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Integration as the reverse of differentiation	20	
B Definite integrals and the area under a curve	21	
C The area between two curves	24	
D The trapezium rule	21	
E Integrating exponentials, $\frac{1}{x}$ and trigonometric functions	18	
F The reverse chain rule	20	
G Integration by substitution	22	
H Integration by parts	21	
I Integration using partial fractions	25	
J Forming and solving differential equations	23	
Roar-level challenge	5	
Total	220	



Section A Integration as the reverse of differentiation

20 marks

1. Find $\int (6x^2 + 4x - 5) dx$. [2]

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2. Find $\int \left(4\sqrt{x} - \frac{3}{x^2} \right) dx$. [3]

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3. Find $\int \frac{(x^2 + 2)^2}{x^2} dx$. [3]

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4. A curve has gradient function $\frac{dy}{dx} = 6\sqrt{x} - \frac{2}{x^2}$, and passes through the point (4, 30). [4]
Find an equation of the curve.

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5. $f'(x) = \frac{2x^2 - 3}{\sqrt{x}}$, $x > 0$.

(a) Write $f'(x)$ in the form $2x^p - 3x^q$, where p and q are constants.

[2]

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(b) Given that the curve $y = f(x)$ passes through the point $(1, 2)$, find $f(x)$.

[4]

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(c) Find $f(4)$.

[2]

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(Total for Question 5 is 8 marks)



Section B Definite integrals and the area under a curve

21 marks

6. Evaluate $\int_1^3 (2x + 1) dx$. [2]

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7. Evaluate $\int_1^4 (3\sqrt{x} - 1) dx$. [3]

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8. Find the area of the finite region enclosed by the curve $y = 4 - x^2$ and the x -axis. [3]

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9. The curve $y = x(x - 2)(x - 3)$ crosses the x -axis at $x = 0$, $x = 2$ and $x = 3$. [5]
Find the total area of the regions enclosed between the curve and the x -axis for $0 \leq x \leq 3$.

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10. (a) Given that $k > 1$ and [4]
$$\int_1^k (6x - 2) \, dx = 20,$$

find the value of k .

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- (b) Find the area of the finite region bounded by the curve $y = (x - 1)(x - 5)$ and the x -axis. [4]
Explain why your integral gives a negative value.

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(Total for Question 10 is 8 marks)



Section C The area between two curves

24 marks

11. Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x$. [3]

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12. Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 4x$. [4]

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13. Find the area of the finite region enclosed by the curves $y = x^2 - 2x$ and $y = 2x - x^2$. [4]

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14. The curve $y = x^3$ and the line $y = x$ enclose two finite regions.
Find the total area of these two regions.

[4]

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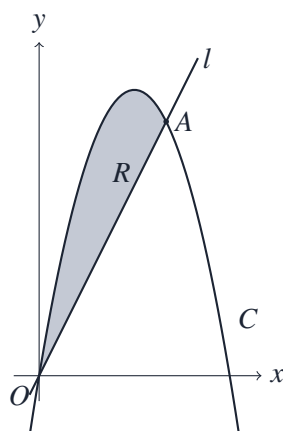
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15. The figure shows the curve C with equation $y = x(6 - x)$ and the line l with equation $y = 2x$. The line meets the curve at the origin O and at the point A . The shaded region R is bounded by C and l .



- (a) Find the coordinates of A . [2]

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- (b) Find $\int_0^4 x(6 - x) \, dx$. [4]

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- (c) Hence find the exact area of R . [3]

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(Total for Question 15 is 9 marks)



Section D The trapezium rule

21 marks

16. Use the trapezium rule with 4 strips to estimate $\int_0^2 \sqrt{1+x^2} dx$, giving your answer to 3 significant figures. [3]

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17. The table gives values of y for certain values of x . [2]

x	0	1	2	3	4
y	2	3.1	4.5	6.2	8

Use the trapezium rule with all the values in the table to estimate $\int_0^4 y dx$.

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18. (a) Use the trapezium rule with 4 strips to estimate $\int_1^3 \ln x dx$, giving your answer to 3 significant figures. [4]

- (b) Given that the exact value is $3 \ln 3 - 2$, state, with a reason, whether your estimate is an overestimate or an underestimate.

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19. (a) Use the trapezium rule with 4 strips to estimate $\int_0^1 e^{x^2} dx$, giving your answer to 3 significant figures. [4]

(b) By considering the shape of the curve $y = e^{x^2}$, explain whether your answer is an overestimate or an underestimate.

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20. The curve C has equation $y = \frac{1}{1 + \sqrt{x}}$, $x \geq 0$.

(a) Complete the table, giving values to 4 decimal places. [1]

x	0	1	2	3	4
y	1	0.5		0.3660	

(b) Use the trapezium rule with all the values in the table to estimate $\int_0^4 \frac{1}{1 + \sqrt{x}} dx$, giving your answer to 3 significant figures. [3]

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(c) State how the accuracy of your estimate could be improved. [2]

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(d) Using your answer to part (b), estimate $\int_0^4 \left(\frac{2}{1 + \sqrt{x}} + 3 \right) dx$. [2]

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(Total for Question 20 is 8 marks)



Section E Integrating exponentials, $\frac{1}{x}$ and trigonometric functions

18 marks

21. Find $\int \left(e^{3x} + \frac{2}{x} \right) dx$. [2]

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22. Find $\int (\cos 2x - 4 \sin x) dx$. [2]

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23. Find the exact value of $\int_0^{\frac{\pi}{2}} (\sin x + \cos 3x) dx$. [3]

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24. Find the exact value of $\int_1^e \left(2x + \frac{3}{x} \right) dx$. [3]

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25. (a) Use the double-angle formula for $\cos 2x$ to show that $\sin^2 x \equiv \frac{1}{2}(1 - \cos 2x)$. [2]

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(b) Hence find the exact value of $\int_0^\pi \sin^2 x \, dx$. [3]

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(c) Find $\int (e^{2x} + e^{-2x})^2 \, dx$. [3]

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(Total for Question 25 is 8 marks)



Section F The reverse chain rule

20 marks

26. Find $\int (2x + 3)^5 dx$. [2]

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27. Find $\int x(x^2 - 4)^3 dx$. [3]

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28. Find $\int \frac{x}{x^2 + 1} dx$. [3]

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29. Find the exact value of $\int_0^{\frac{\pi}{3}} \sin x \cos^3 x dx$. [4]

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30. (a) By writing $\tan x = \frac{\sin x}{\cos x}$, show that $\int \tan x \, dx = \ln |\sec x| + C$. [3]

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(b) Hence find the exact value of $\int_0^{\frac{\pi}{3}} \tan x \, dx$. [3]

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(c) Find $\int \frac{4x}{(x^2 + 3)^2} \, dx$. [2]

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(Total for Question 30 is 8 marks)



Section G Integration by substitution

22 marks

31. Use the substitution $u = 4x + 1$ to find $\int \sqrt{4x + 1} \, dx$. [3]

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32. Use the substitution $u = x - 1$ to find $\int x(x - 1)^5 \, dx$. [4]

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33. Use the substitution $u = x - 1$ to find the exact value of $\int_1^5 x\sqrt{x - 1} \, dx$. [4]

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34. Use the substitution $u = \sqrt{x}$ to show that

[4]

$$\int \frac{1}{x + \sqrt{x}} dx = 2 \ln(1 + \sqrt{x}) + C.$$

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35. (a) Using the substitution $u = 2x - 1$, show that

[4]

$$\int_1^5 \frac{x}{\sqrt{2x-1}} dx = \frac{1}{4} \int_1^9 \left(u^{\frac{1}{2}} + u^{-\frac{1}{2}} \right) du.$$

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(b) Hence find the exact value of $\int_1^5 \frac{x}{\sqrt{2x-1}} dx$.

[3]

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(Total for Question 35 is 7 marks)



Section H Integration by parts

21 marks

36. Find $\int x \cos x \, dx$. [3]

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37. Find $\int x e^{2x} \, dx$. [3]

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38. By writing $\ln x$ as $1 \times \ln x$, use integration by parts to find $\int \ln x \, dx$. [3]

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39. Find the exact value of $\int_1^e x^2 \ln x \, dx$.

[4]

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40. (a) Find $\int x^2 e^x \, dx$.

[5]

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(b) Hence find the exact value of $\int_0^1 x^2 e^x \, dx$.

[3]

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(Total for Question 40 is 8 marks)



Section I Integration using partial fractions

25 marks

41. Express $\frac{3}{(x-1)(x+2)}$ in partial fractions, and hence find $\int \frac{3}{(x-1)(x+2)} dx$. [4]

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42. Given that $\frac{5x+1}{(x+1)(x-1)} \equiv \frac{2}{x+1} + \frac{3}{x-1}$, find $\int \frac{5x+1}{(x+1)(x-1)} dx$. [3]

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43. Show that [5]

$$\int_2^4 \frac{x+5}{(x-1)(x+2)} dx = \ln 6.$$

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44. (a) Show that $\frac{2x^2 + 3}{x^2 - 1} \equiv 2 + \frac{5}{x^2 - 1}$.

[5]

(b) Hence find $\int \frac{2x^2 + 3}{x^2 - 1} dx, x > 1$.

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45. $f(x) = \frac{x^2 + 7x + 1}{(x-1)^2(x+2)}, \quad x > 1.$

- (a) Express $f(x)$ in the form $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$. [3]

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- (b) Hence show that $\int_2^4 f(x) \, dx = a + \ln b$, where a and b are integers to be found. [5]

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(Total for Question 45 is 8 marks)



Section J Forming and solving differential equations

23 marks

46. Solve the differential equation $\frac{dy}{dx} = 2y$, given that $y = 3$ when $x = 0$. [3]

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47. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{x}{y}$. [3]

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48. Solve the differential equation $\frac{dy}{dx} = y \cos x$, given that $y = 2$ when $x = 0$. [4]
Give your answer in the form $y = f(x)$.

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49. The mass, m grams, of a substance decreases at a rate proportional to its mass.
Initially the mass is 50 g, and after 5 hours it is 40 g.

[5]

- (a) Write down a differential equation relating m and the time t hours.
- (b) Solve the differential equation to find m in terms of t .
- (c) Find the time taken for the mass to fall to 20 g.

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50. Liquid drains from a container. The volume, $V \text{ cm}^3$, of liquid in the container at time t minutes satisfies

$$\frac{dV}{dt} = -k\sqrt{V},$$

where k is a positive constant. Initially $V = 100$, and after 10 minutes $V = 64$.

(a) Solve the differential equation to show that $\sqrt{V} = 10 - 0.2t$. [5]

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(b) Find the time taken for the container to empty. [2]

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(c) Find the volume of liquid in the container after 20 minutes. [1]

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(Total for Question 50 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

51. Use integration by parts twice to find $\int e^x \sin x \, dx$.

[5]

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END OF WORKSHEET

Total: 220 marks